

THE DEVELOPMENT BACKLOG · AUTHORED IN THE PASS-5 SPIRIT

What the factory builds **next**.

EVALS FIRST — EVEN HERE. — every item names the gate that will prove it.

Eight epics across three horizons, every item carrying an **effort tier**, a **priority**, a **definition of done**, and — in the house discipline — the **evidence that put it here**. Nothing below is a wish: each entry traces to a measured failure, a named roadmap commitment, a release gate already written down, or a positioning bet made explicit. When an item is scheduled, it decomposes into sealed Task-Specs and these “proves” lines become the evals.

8 EPICS

31 ITEMS

EVIDENCE-CITED

GATES PROPOSED PER ITEM

01 **How to read this. Priorities, tiers, and what “proves” means.**

6

P0 ITEMS

9

P1 ITEMS

11

P2 ITEMS

5

P3 ITEMS

3

HORIZONS

31

DEFINED GATES

PRIORITY SCALE

P0 the thesis is at risk — the project is currently contradicting its own claims
· **P1** the roadmap's named commitments · **P2** compounding value, scheduled after the commitments · **P3** breadth that can wait for pull.

EFFORT TIERS — THE SIX-TIER SCALE

The same scale the gate enforces on Task-Specs: **XS** ≤ 1 write path · **S** ≤ 2 · **M** ≤ 3 · **L** ≤ 5, one coherent goal · **XL / XXL** are **nodes** — never runnable, they dispatch children.

PROVES — PROPOSED GATES

Each item ends in the check that will decide it. Existing tokens (**DOCTOR_HOST_TESTS=PASS**) are the real gates today; *proposed* tokens (**EVAL_GATE=OK**) become real when the item's Task-Spec is authored and sealed. A proposal is not a verdict.

Why Epic A outranks everything. Converge's pitch is “done is proven, never claimed” — and as of 2026-08-05 its own proof surface is dark: every CI run since **2026-08-03T20:17Z** has failed in 3–13 seconds with no step executed and job logs returning 404 (runner-allocation shape, not a test failure), while the README still says “CI is public and green.” Eight commits of new gate logic sit unverified by the gauntlet. A trust product wearing a red badge under a green claim is the one contradiction the thesis cannot survive — so relighting the floor is the whole of horizon one.

02 The three horizons. Restore the floor → move settlement server-side → compound.

NOW · NEXT · LATER

epics by horizon – each unlocks the next



NOW — 0.1.x

Relight the proof surface. Every claim the project makes must be measurable again before anything new ships. Six items, all small, all P0.

LATER — 0.3+

Compound. Distribution, the Cockpit's four release gates, deeper verification, wider platforms — value that stacks on a green floor.

NEXT — 0.2

Settlement goes server-side. The two roadmap commitments (eval-gate, Manager) plus the interop bet that positions FAST as the front door.

03

Blocked-by, item level

same relation Pass 6 mirrors onto a board

Reading the graph: A1 gates the whole machine half of this backlog — an eval-gate cannot ship into a CI that does not execute, and the Manager rides on both. The Cockpit chain (F1-F4 → F5) is the release contract already written in `apps/cockpit/PROVING-GROUNDS.md`: the app ships in the package only when all four cases have exercised the same WorkspaceSnapshot 3.0 contract.

A Relight the proof surface — the claims must be measurable again.

HORIZON · NOW (0.1.x)

6 ITEMS

ALL P0

EXIT · GAUNTLET GREEN + CLAIMS TRUE

A1 P0 S infra

Diagnose and fix the CI execution outage

Every run since `2026-08-03T20:17Z` — across `main`, `feat/e2e`, and PR branches — completes in 3–13 s with all three jobs failed before any step ran, and job logs 404. Healthy runs took 500–730 s. That shape points at runner allocation, Actions billing/quota, or org policy — not at the tests. Diagnose in Actions settings, fix, then re-run the gauntlet on `main`.

EVIDENCE

GitHub Actions API, 2026-08-05: 14 failures / 3 cancelled in the last 30 runs; last green `ci.yml` run 605 s on 2026-08-03.

PROVES

Two consecutive `ci.yml` runs on `main` conclude `success` on both legs (ubuntu, macos) with normal durations; the README badge renders green.

A2 P0 XS test-portability

Make the doctor-host suite portable to apt-installed shellcheck

The suite synthesizes “shellcheck absent” by dropping only the homebrew dir from PATH. On hosts where shellcheck lives in `/usr/bin` — exactly what the CI Linux leg installs — the tool stays visible, rows [2]–[3] report 4 FAILs, and the suite lands `DOCTOR_HOST_TESTS=FAIL` against correct product behavior. Row [4] already SKIPs honestly in the same situation; reuse the suite's own `mirror_without` technique (built for python3) for shellcheck.

EVIDENCE

Suite run 2026-08-05 on Linux with `/usr/bin/shellcheck`: rows 13, failed 4.

PROVES

`DOCTOR_HOST_TESTS=PASS` on a host with apt shellcheck *and* on a brew host; absence rows SKIP — never FAIL — when absence cannot be synthesized.

A3 P0 XS test-deps

Guard the hmac-envelope suite's xxd dependency

`skills/task-spec/tests/test-hmac-envelope.sh` hard-calls `xxd` (line 77) with no presence check — on a host without it the suite dies rc=127 with a bare *command not found*. Every other suite in the package either guards its deps or declares an honest UNRUNNABLE. Use an `od / printf` fallback or a preflight check.

EVIDENCE

Suite run 2026-08-05: rc=127 at line 77 on a minimal container.

PROVES

The suite exits with its own token on xxd-less hosts — `PASS` via fallback, or a named UNRUNNABLE — and never a bare 127.

A5 P0 S hygiene

Take the binary weight out of the repository

`converge-brand-kit-v1.zip` (2.2 MB) is committed at the root and duplicates the extracted `brand-kit-v1/` beside it; `brand/` carries 13 MB and `docs/` 17 MB of binaries around a ~300 KB product; `.git` is 44 MB. Move heavy assets to release attachments or LFS, delete the duplicate zip, and keep the npm `files:` allowlist as the slim boundary it already is.

EVIDENCE

`du` audit 2026-08-05.

PROVES

No committed archive duplicates a committed tree; a fresh clone lands under an agreed budget; CI's removed-asset needle check stays green.

A4 P0 XS cockpit-test

Fix the Cockpit agent-registry test's USER assumption

`server/agent-registry.test.mjs` builds its ACP environment from `process.env` and throws *“environment USER must be a non-empty bounded string”* when `USER` is unset — common in containers and CI images. The production path (`server/index.mjs`) already filters empty vars correctly; the fixture should pass an explicit environment.

EVIDENCE

`npm run cockpit:check` 2026-08-05: 36/37 with USER unset; 37/37 with USER set.

PROVES

`cockpit:check` is green under `env -u USER`.

A6 P0 XS honesty

Refresh the numbers the docs claim

The README and the deck say **52** hermetic loop-kernel checks; the suite now proves **57**. Small, but this project's whole brand is that its claims carry receipts — stale numbers are the cheap version of the failure it exists to prevent. Sweep README, deck, and these documents after A1–A4 land.

EVIDENCE

`test-loop-kernel.sh` 2026-08-05: “PASS — 57 loop-kernel checks green.”

PROVES

Documented counts match a fresh local run's output; ideally a claims-lint joins the version-unity family so drift fails the build.

B The CI eval-gate — settlement moves to where merges happen.

HORIZON · NEXT (0.2)

ROADMAP COMMITMENT

3 ITEMS

EXIT · A PR CANNOT MERGE ON A CLAIM

The roadmap names it: “*server-side re-verification of every green claim.*” Today the referee lives on laptops; a settlement layer that only runs client-side is advice, not enforcement. This epic puts the gate in the merge path — and it is the single highest-leverage strategic move in the backlog: it turns Converge from a method into infrastructure.

B1 P0 M action

The re-verification GitHub Action

On every PR touching `cvg/tasks/` or claiming settlement: re-verify the HMAC seal against the repo key, re-run the changed specs' evals in the runner, and publish one check with one token. The stub attack — a faked spec with hand-edited evals — must land RED *in CI*, not just locally.

EVIDENCE

README §Status names the CI eval-gate as deliberately absent from 0.1.0; deck p27 lists it as roadmap.

PROVES

Proposed `EVAL_GATE=OK|FAIL` as a required check; a seeded stub-attack fixture PR shows the check RED.

B2 P1 XS docs

The branch-protection recipe

Document the Git-native settlement pattern the FAQ already sketches: Actions as scheduler, the PR as settlement, branch protection as the gate — with the eval-gate check required. One page, copy-pasteable.

EVIDENCE

README FAQ “What's the Manager?” names this trio as the intended orchestration substrate.

PROVES

The converge repo itself enforces the recipe; the doc shows the exact protection settings.

B3 P1 S action

The receipt & provenance verifier

Extend the gate to the rest of the trust chain: verify the receipt hash chain (`cvg.execution-receipt.v1` pins spec sha, profile sha, eval digest) and re-hash Pass-4 provenance on PRs that touch plans — the same drift check `cvg review --check` runs, now server-side.

EVIDENCE

Receipt format and provenance re-hash are shipped client-side (Complete Edition §14).

PROVES

A PR that edits a settled receipt, or a plan after its review, goes RED with the drift named.



The Manager — fleet dispatch across the ready frontier.

HORIZON · NEXT (0.2)

ROADMAP · TOP ITEM

2 ITEMS · 1 NODE

EXIT · N TASKS, N NAMED LANDINGS

The loop is single-task by design today; the Manager decides *which* task runs, when, in parallel, watching PRs. The design stance is already set — a Git-native world provides most of the orchestration (Actions as scheduler, PR as settlement) — so the Manager schedules **around** the loop rather than replacing it. Its safety story is inherited, not invented: every child is a normal bounded loop under a sealed spec and the write fence.

C1

P1

XL · NODE

orchestration

Frontier dispatch — the node

Never runnable; dispatches three children when scheduled: **C1a** the scheduler (consume `cvg ready`, cap concurrency, one loop per task in its own worktree) · **C1b** the locks (reuse `cvgt transition`'s locked, logged moves so two dispatchers cannot claim one task) · **C1c** the steward (watch each opened PR; on CI red, requeue a bounded fix loop against the same sealed spec).

EVIDENCE

README §Status: “the Manager ... is next on the roadmap”; FAQ commits to the Git-native shape.

PROVES

Proposed `MANAGER=DISPATCHED|IDLE`; N ready tasks yield N terminal states with board parity intact and zero fence violations — proven first with stub engines, in the loop-kernel style.

C2

P2

M

cockpit

The fleet view in Cockpit

Surface the dispatch frontier and each child loop's live state (iteration, strikes, budget burn) in the existing Runs surface — still read-only, still one snapshot door, still zero frontend-derived truth.

EVIDENCE

Cockpit's Runs surface and SSE bridge already render single-loop attempts.

PROVES

Snapshot schema gains a versioned fleet block; the existing snapshot suite proves emitted = checked-in.

D Spec-layer interop — ingest, don't compete.

HORIZON · NEXT (0.2)

STRATEGIC BET

2 ITEMS

EXIT · OTHER SPECS COMPILE INTO SEALED SPECS

The spec-authoring layer is crowded and commoditizing; Converge's moat is the settlement layer beneath it — the seal, the fence, the bounded loop, the cross-family refute. So interoperate upward: accept what Spec Kit or Kiro produce as *input*, and let Converge be the layer that makes any of them enforceable.

D1 P1 M cli

cvg tasks import — compile external specs to Task-Specs

Map an external spec's tasks onto scaffolded Task-Specs: goals and paths carried over, **evals left TODO and the spec born** **status: blocked** until a human authors and seals them. The trust boundary is explicit: import accelerates authoring; it never auto-signs.

EVIDENCE

Positioning review 2026-08-05: settlement is the defensible layer; authoring is commoditized.

PROVES

Proposed **IMPORT=OK** ; imported specs fail **tasks gate** until evals exist — proven by a fixture round-trip suite.

D2 P1 XS positioning

Make the FAST lane the front door

Reorder the story: README and quickstart lead with **cvg lane** + FAST (task → bind → loop on an existing repo, minutes to first sealed settlement); the full nine-pass descent is positioned as the high-assurance path for new systems and regulated work — not the default ask.

EVIDENCE

The quickstart already runs FAST-shaped; ceremony-first framing is the main adoption headwind.

PROVES

A newcomer path measured in minutes-to-first- **TASK_LOOP=SETTLED** on a toy repo, documented and CI-exercised via the clean-room suite.

E Distribution & trust — easier to get, harder to fake.

HORIZON · LATER (0.3+)

4 ITEMS

EXIT · SIGNED, PUBLISHED, JOINABLE

E1 P2 S packaging

Publish to the npm registry proper

Today the npm door is `github:` installs only. Publish the versioned unit to the registry (scoped name, provenance attestation), keeping the `files:` allowlist as the boundary.

EVIDENCE

package.json installs via github: URL; no registry entry.

PROVES

`npm i -g` from the registry → `INSTALL=OK`; version-unity holds across the published artifact.

E2 P2 S docs

Docs site on Pages

Serve the interactive set — the presentation walkthroughs, the Complete Edition v2, and this backlog — from GitHub Pages, so the canonical documents are a URL, not a checkout.

EVIDENCE

All documents are standalone HTML already; only hosting is missing.

PROVES

A Pages workflow publishes on release; links in README resolve.

E3 P2 M crypto

Per-author signatures — the named upgrade

The trust ladder is honest about symmetric HMAC: tamper-evident, not tamper-proof, binding “a repo-key holder stamped this.” Ship the named upgrade — Ed25519/DSSE per-author signatures — as an opt-in Tier-1 strengthening, plus signed release artifacts.

EVIDENCE

Complete Edition §14: “per-author signatures (Ed25519/DSSE) are the named upgrade.”

PROVES

The gate verifies either seal form; a DSSE-signed spec attributes its signer; releases carry verifiable signatures.

E4 P2 XS community

The contribution surface

CONTRIBUTING.md (the gauntlet is the review bar), issue and PR templates that ask for tokens-not-claims, and a handful of good first issues — this backlog has several.

EVIDENCE

Single-maintainer repo; no contribution scaffolding yet.

PROVES

Templates live under `.github/`; a first external PR lands through the gauntlet.

F Cockpit release gates — the contract PROVING-GROUNDS.md already wrote.

HORIZON · LATER (0.3+)

5 ITEMS

EXIT · ALL FOUR CASES ON SNAPSHOT 3.0

The Cockpit's own file states the bar: *"The application is release-ready only when all four cases have exercised the same WorkspaceSnapshot 3.0 contract without frontend-derived truth."* These items are that sentence, made schedulable.

F1 P2 S proving-ground

Rerun uc-01 against WorkspaceSnapshot 3.0

The prior V0 observation is not current release evidence. Rerun the analytics-engineering FULL-lane case against the 3.0 contract; Cockpit shows whatever the fresh CLI snapshot reports — it must not synthesize lifecycle data.

EVIDENCE

`apps/cockpit/PROVING-GROUNDS.md`: gate "Open — dated rerun and owner review required."

PROVES

A dated rerun + owner review recorded in PROVING-GROUNDS.md; the gate row flips.

F2 P2 M proving-ground

Stand up uc-02 / uc-03 / uc-04

Data-engineering (NORMAL), software-engineering (NORMAL), and AI-engineering (FAST + failure paths) proving workspaces — the failure-path case matters most: the Cockpit must render BLOCKED and REFUTED as first-class, not as gaps.

EVIDENCE

PROVING-GROUNDS.md: three cases "Not run — no proving-ground workspace present."

PROVES

Three dated case runs on Snapshot 3.0, each with owner review recorded.

F3 P2 S ask-path

The authorized provider smoke test

One explicitly authorized provider turn through Ask, with before/after workspace hashes proving read-only held, and owner review of the provider-data and retention boundary.

EVIDENCE

PROVING-GROUNDS.md names this as the missing sign-off beside the hermetic fake-agent suites.

PROVES

Before-hash = after-hash across the turn; the review is recorded.

F4 P2 S upstream

Unblock Codex in Ask — adapter isolation

Codex stays statically blocked before spawn because the pinned ACP adapter cannot suppress its local read/search tools. Track the upstream adapter; lift the block only when context isolation is provable in the fake-agent suite.

EVIDENCE

README + Cockpit docs: "Codex choice visible but blocked pending adapter isolation."

PROVES

The isolation test that today proves the block instead proves the suppression; the static block is removed in the same change.

F5 P2 XS release

Ship Cockpit in the package

When F1–F4 are green: include the Cockpit in the published unit and retire the one-release `control-room:*` aliases — restating the dependency claim honestly (the CLI stays zero-dependency; the Cockpit is the optional Node surface).

EVIDENCE

README: "not included ... while the live proving-ground cases are still being completed."

PROVES

The package ships it; README's claim and `files:` agree; blocked-by edges F1–F4 all closed.

G Verification depth — close the gaps the docs already admit.

HORIZON · LATER (0.3+)

4 ITEMS

EXIT · THE HONEST FOOTNOTES RETIRE

G1 P2 M portability

cvgt lint on bash 3.2 — close the declared gap

The CHANGELOG says it plainly: the lint uses associative arrays, the portability floor is 3.2, and “the real fix is a bash-3.2 rewrite or dropping the floor.” Rewrite the cycle/overlap checks floor-clean so stock macOS gets a real verdict instead of `LINT=UNSUPPORTED`.

EVIDENCE

CHANGELOG [Unreleased] \$Documented: “The gap itself is open.”

PROVES

`LINT=OK|WARN|ISSUES` on the macOS 3.2 CI leg; UNSUPPORTED retires from the token grammar.

G2 P2 S loop

Close the worktree resume gap

The canonical presentation carries the caveat: “default worktree resume gap may restart attempt 1.” A resumed run must continue at attempt N+1 with its strike counter and fingerprint intact — anything else quietly refunds spent budget.

EVIDENCE

`presentation/converge.html` §12, resume command annotation.

PROVES

A loop-kernel row: kill at attempt 2, `--resume`, observe attempt 3 with state.env continuity — in worktree mode.

G3 P2 S holdout

Holdout authoring aids

The holdout is the highest-leverage anti-reward-hacking device in the chain, and today it is hand-rolled. Ship templates and an authoring checklist in the task-spec skill, and a gate WARN when a High-blast-radius spec carries no `Holdout`.

EVIDENCE

The REFUTED production case was caught by a holdout; blast-radius rules are already computed.

PROVES

The WARN fires on a High spec without holdout in the fixture suite; templates ship in the skill's references.

G4 P3 M judging

The judge panel — diversity over redundancy

`cvx verify --panel`: N judges with distinct lenses (correctness, security, does-it-reproduce), refutation-majority semantics, per-lens verdicts recorded in the receipt. One judge catches one failure mode; a diverse panel catches the ones redundancy can't.

EVIDENCE

Single-judge tier-2 is shipped; 2026 verification literature favors lens-diverse refutation.

PROVES

Panel verdicts recorded per lens; a seeded plausible-but-wrong fixture is REFUTED by the panel where the single judge upholds.

H Platform breadth — more engines, more boards, more hosts.

HORIZON · LATER (0.3+)

3 ITEMS

EXIT · WIDER, SAME GATES

H1 P3 S engine

Gemini engine adapter

One file, two calls — `--available` and `--prompt-file` — and the cross-family pool for Pass 4 and tier-2 grows by a vendor. Engines are adapters by construction; this is the proof by example.

EVIDENCE

Pass-4 dispatch already names Gemini as an adversary family; no adapter ships.

PROVES

`cvg doctor` counts family *google*; the loop settles a stub task via the new adapter file, kernel unchanged.

H2 P3 M tracker

Jira adapter to parity

Bring the Jira adapter through the same six-verb, fake-backed parity suite that proves Linear and GitHub offline — 145 checks, no network, no credentials.

EVIDENCE

Register names Linear / GitHub / Jira; the offline parity suite is the bar.

PROVES

The register suite passes with the Jira adapter under the fake, verb for verb.

H3 P3 L host

Windows beyond Git Bash

Today Windows means Git Bash or WSL, documented honestly. Decide the native story deliberately — a supported PowerShell shim for the referee, or a documented WSL-first stance — rather than by silence.

EVIDENCE

README requirements: “run everything in Git Bash or WSL.”

PROVES

A decision record (ADR) plus, if native: doctor-host probes for the Windows toolchain and a CI leg to hold it.

04 The master table. Every item, one line each.

ID	ITEM	EFFORT	PRIORITY	HORIZON	BLOCKED BY
A1	CI execution outage — diagnose & fix	S	P0	NOW	—
A2	doctor-host suite portability (apt shellcheck)	XS	P0	NOW	—
A3	hmac-envelope xxd guard	XS	P0	NOW	—
A4	Cockpit agent-registry USER fixture	XS	P0	NOW	—
A5	Repo binary weight out (zip dup, heavy assets)	S	P0	NOW	—
A6	Claims refresh (52 → 57 et al.)	XS	P0	NOW	A1–A4
B1	Re-verification GitHub Action (the eval-gate)	M	P0	NEXT	A1
B2	Branch-protection recipe	XS	P1	NEXT	B1
B3	Receipt & provenance verifier	S	P1	NEXT	B1
C1	Frontier dispatch — node (C1a scheduler · C1b locks · C1c steward)	XL	P1	NEXT	B1
C2	Fleet view in Cockpit	M	P2	NEXT	C1
D1	<code>cvg tasks import</code> — external specs in	M	P1	NEXT	—
D2	FAST lane as the front door	XS	P1	NEXT	—
E1	npm registry publish + provenance	S	P2	LATER	—
E2	Docs site on Pages	S	P2	LATER	—

ID	ITEM	EFFORT	PRIORITY	HORIZON	BLOCKED BY
E3	Ed25519/DSSE per-author signatures + signed releases	M	P2	LATER	—
E4	Contribution surface	XS	P2	LATER	—
F1	uc-01 rerun on Snapshot 3.0	S	P2	LATER	A4
F2	uc-02 / uc-03 / uc-04 proving workspaces	M	P2	LATER	—
F3	Authorized provider smoke test	S	P2	LATER	—
F4	Codex ACP isolation unblock	S	P2	LATER	upstream
F5	Ship Cockpit in the package	XS	P2	LATER	F1-F4
G1	<code>cvg lint</code> bash-3.2 rewrite	M	P2	LATER	—
G2	Worktree resume gap	S	P2	LATER	—
G3	Holdout authoring aids + WARN	S	P2	LATER	—
G4	The judge panel (lens diversity)	M	P3	LATER	—
H1	Gemini engine adapter	S	P3	LATER	—
H2	Jira adapter to parity	M	P3	LATER	—
H3	Windows beyond Git Bash	L	P3	LATER	—

C1's three children (C1a–C1c) are counted inside the node, Task-Spec style — an XL is never runnable and only its children dispatch. 31 items total: 28 rows here plus the three children the node will cut.

05 **Backlog → Task-Specs.** This document is upstream of Pass 5, on purpose.

This backlog sits at plan altitude — the same altitude as swimlane legs: one responsibility, a proves-line, no eval bodies. Scheduling an item means walking it through the factory's own front half, not around it:

```
$ cvg lane "fix the doctor-host suite portability"      # → FAST
$ cvg tasks new fix-doctor-host-portability XS         # scaffold – evals FIRST
# author the evals from this item's "proves" line, then:
$ cvg tasks validate cvg/tasks/T-*.md
$ cvg tasks gate --stamp cvg/tasks/T-*.md              VERDICT: DELEGATE · TIER=1
$ cvg bind --task cvg/tasks/T-*.md && cvg loop --issue T-... --agent codex
TASK_LOOP=SETTLED                                     # the item leaves this page
```

THE EXIT CRITERION FOR THIS WHOLE DOCUMENT

Every P0 item settled through the loop with its token green, the eval-gate required on `main`, and the README's status section describing a state that a stranger can re-measure. At that point this backlog regenerates from the board — it stops being a file and becomes a query.

PROVENANCE OF THIS BACKLOG

Compiled 2026-08-05 from: the GitHub Actions API (A1), local runs of all 22 suites (A2–A4, A6), a du audit (A5), README §Status and FAQ (B, C), `apps/cockpit/PROVING-GROUNDS.md` (F), CHANGELOG [Unreleased] (G1), `presentation/converge.html` §12 (G2), the Complete Edition's trust ladder (E3), and the 2026-08-05 positioning review (D). Where an item's need was measured, the measurement is quoted on the item.

CONVERGE · THE BACKLOG · v0.1.0 → 0.3 – evals first, even here · every item names the gate that will prove it

- The Complete Edition v2
- Method walkthrough
- Task-Spec